

MUHAMMAD ADEEL SAJJAD

SYSTEMS, INFRASTRUCTURE & CYBER-PHYSICAL SECURITY ENGINEER · B.SC. ELECTRICAL ENGINEERING — P

Email: eng.adeelhaider@gmail.com | GitHub: <https://github.com/Haidriyam> | Location: Dubai, UAE
GPG Fingerprint: 9E4F 8C2B 117A 3D90 5F1E 8A24 C37B 4290 88DF 10E5

1. PROFILE / SUMMARY

Systems & Infrastructure Engineer with a foundational degree in Electrical Engineering (Power Systems) and over 8 years of hands-on enterprise infrastructure experience. Rigorous training in electromagnetic field modeling, resonant power electronics, and high-voltage transmission, combined with deep operational expertise in deterministic kernel telemetry pipelines, bare-metal infrastructure automation, and industrial control systems (SCADA) network boundary defense.

2. RESEARCH INTERESTS

- **Resilient Cyber-Physical Systems (CPS):**
Failure modes & attack surfaces where unauthenticated industrial protocols (Modbus-TCP, DNP3) couple with physical power converter dynamics and protection relays.
- **Smart Grid Security & Anomaly Detection:**
Deterministic packet inspection, state-estimation defense against false data injection (FDI), and IEC 62443 / Purdue reference architecture enforcement.
- **Resonant Inductive Coupling:**
High-Q magnetic resonance link optimization, mutual inductance parameter extraction, and zero-voltage-switching (ZVS) power electronics.
- **Distributed Testbed Infrastructure:**
Hardware-in-the-loop (HIL) environments, kernel-level zero-drop ring buffering, and reproducible bare-metal infrastructure orchestration.

3. EDUCATION

Bachelor of Science in Electrical Engineering — Power Systems

Graduated 2016

The Islamia University of Bahawalpur · Pakistan

Specialization: Power Systems & Electrical Infrastructure | Academic Standing / Status: Degree Conferred (2016)

Capstone Thesis: "Resonant Inductive Wireless Power Transmission System"

Modeled Maxwell's equations and coupled-mode theory for near-field electromagnetic energy transfer over variable air gaps. Designed, wound, and tuned helical magnetic induction coils. Developed primary and secondary resonant LC tanks to cancel reactive power and maximize power
Supervisor / Advisor: undefined

4. ENGINEERING EXPERIENCE

Senior IT & Infrastructure Systems Manager

Nov 2025 — Present

Urban 411 Design Products Trading · Systems & Infrastructure Operations — Dubai, UAE

Directing enterprise IT and cloud systems infrastructure with complete lifecycle management across multi-tier hosting, automated telemetry pipelines, perimeter hardening, and automated CI/CD deployments.

KEY TECHNICAL CONTRIBUTIONS:

- Achieved 40%+ reduction in server response latency through reverse-proxy caching, database query optimization, and kernel parameter tuning.
- Enforced a 99.8% operational SLA across multi-tier infrastructure by eliminating single points of failure and automating recovery.
- Deployed centralized Wazuh SIEM telemetry pipeline ingesting host events and audit logs with automated threat detection rules.

Systems: Wazuh HIDS/SIEM/XDR Security Platform, Linux Distributions (Ubuntu Server, Debian, RHEL), Nginx & Apache Web Servers, GitHub Actions & Bash CI/CD Workflows, Certbot & Let's Encrypt Automated SSL/TLS, Multi-Tier Cloud Hosting (AWS EC2, Lightsail)

IT & Infrastructure Operations Manager

2020 — 2025

TECDEAL General Trading LLC · Infrastructure & Operations — Dubai, UAE

Led infrastructure architecture and systems operations across distributed commercial hubs, securing local network topologies, gateway firewalls, database clusters, and virtualization environments.

KEY TECHNICAL CONTRIBUTIONS:

- Segmented distributed commercial hub networks using VLANs and gateway ACLs, completely isolating sensitive transactional servers.
- Optimized relational database query execution plans and index trees, eliminating peak-hour query bottlenecks.
- Implemented automated multi-region snapshot failover workflows ensuring rapid recovery in the event of local infrastructure disruption.

Systems: Gateway Firewalls & Multi-VLAN Subnetting, Server Virtualization Environments, Relational Database Clusters (MySQL / PostgreSQL), Multi-Region Cloud Snapshot Pipelines, Hardware Access Control & Bastion Proxies

Systems & Web Development Engineer

2016 — 2019

Techvision · Systems & Web Engineering — Bahawalpur, Pakistan

Managed production Linux LAMP/LEMP server environments, performing routine security audits, system patching, automated shell scripting, and local network maintenance.

KEY TECHNICAL CONTRIBUTIONS:

- Maintained consistent server uptime through automated cron diagnostics, preemptive disk space monitoring, and offsite log archiving.
- Executed systematic security patching schedules across production Linux servers, eliminating exposed service vulnerabilities.
- Designed reliable shell-driven offsite database backup replication routines with automated integrity verification.

Systems: Linux LAMP & LEMP Production Stacks, Apache & Nginx Web Servers, MySQL Database Administration, Bash Shell & Cron Automation Daemons, Physical Routing & Managed Switch Hardware

5. SELECTED PROJECTS & SYSTEMS

[PROJECT / WPT-001] Resonant Inductive Wireless Power Transmission System Aug 2015 — Jun 2016 / CAPSTONE THESIS

Domain: Electrical Engineering / Electromagnetic Modeling | Claim: Modeled Maxwell equations and coupled-mode theory to design, construct, and evaluate. Modeled Maxwell's equations and coupled-mode theory for near-field electromagnetic energy transfer over variable air gaps (0–150 mm). Designed, wound, and tuned helical magnetic induction coils. Developed primary and secondary resonant LC tanks to cancel reactive power and maximize power factor. Fabricated switching driver/inverter hardware with gate-drive isolation. Evaluated transmission efficiency curves, Q factors, and thermal behavior using DSO oscilloscopes, RF power meters, and signal generators. Authored thesis documentation containing mathematical derivations, schematic designs, and empirical performance metrics, defended before the examination board.

Stack: Resonant Induction, LC Tank Matching, Power Electronics, Helical Coils, DSO Metrology, Capstone Thesis

[PROJECT / CPS-001] MICROGRID RESILIENT CONTROL

2025 / OPEN-SOURCE CPS REPOSITORY

Domain: Cyber-Physical Systems / Electrical Infrastructure / Control Security | Claim: Implements secondary frequency consensus restoration on islanded AC microgrids under False Data Injection (FDI) attacks. Features distributed cooperative control algorithms for inverter-based distributed energy resources (DERs), modeling physical frequency droop dynamics coupled with resilient state exchange across communication topologies. Incorporates detection and mitigation of adversarial data corruption targeting consensus communication links.

Stack: Microgrid, Resilient Control, False Data Injection, Consensus Algorithms, Inverter Droop, Cyber-Physical

[PROJECT / CPS-002] BATTERY EKF STATE ESTIMATOR

2025 / OPEN-SOURCE CPS REPOSITORY

Domain: Cyber-Physical Systems / Battery Systems / Embedded Security | Claim: Pairs discrete Extended Kalman Filtering with strict CAN-bus frame boundary Discrete Extended Kalman Filter (EKF) for Li-ion State-of-Charge estimation combined with boundary-checked CAN-bus telemetry parsing and frame fuzzing. Implements an equivalent circuit model (Thevenin ECM) capturing battery polarization dynamics and nonlinear open-circuit voltage curves. Pairs filtering mathematics with defensive embedded software engineering, including frame validation, range boundaries, and fuzzing resistance.

Stack: Battery Systems, Extended Kalman Filter, State of Charge, CAN-Bus, Fuzzing, Embedded Security

[PROJECT / CPS-003] SCADA IDS TELEMETRY ENGINE

2025 / OPEN-SOURCE CPS REPOSITORY

Domain: Industrial Security / SCADA / OT Security | Claim: Performs deep APDU protocol inspection and zero-trust command whitelisting to detect industrial Deep-packet APDU dissector for Modbus TCP/IEC SCADA protocols with zero-trust command whitelisting and substation physical anomaly detection. Inspects industrial network traffic at the application protocol data unit (APDU) layer, enforcing zero-trust command policies (e.g. blocking unauthorized coil writes or register overrides) and verifying telemetry against physical domain constraints.

Stack: SCADA, Modbus TCP, APDU Inspection, OT Security, Industrial IDS, Zero-Trust

[PROJECT / WPT-002] WIRELESS POWER TRANSMISSION SIMULATION

2025 / OPEN-SOURCE NUMERICAL SUITE

Domain: Electrical Engineering / Electromagnetic Modeling | Claim: Provides numerical validation of resonant inductive wireless power transfer dynamics, Mathematical modeling of resonant magnetic inductive coupling, unloaded Q-factor dynamics, and mutual link efficiency optimization. Developed in Python with automated numerical testing, this repository translates physical electromagnetic hardware experiments into reusable simulation models.

Stack: Wireless Power, Resonant Induction, Numerical Simulation, Coupled-Mode Theory, Q-Factor, Python

[PROJECT / SEC-001] QUANTUM PQC TLS CANARY

2025 / OPEN-SOURCE SECURITY REPOSITORY

Domain: Post-Quantum Cryptography / CPS Security / Network Security | Claim: Demonstrates hybrid post-quantum key encapsulation (ML-KEM-768 + X NIST FIPS 203 ML-KEM-768 + X25519 hybrid key encapsulation canary protecting edge CPS telemetry against Harvest-Now-Decrypt-Later (HNDL) attacks. Explores post-quantum cryptographic primitives on constrained communication channels, measuring overhead, handshake latency, and TLS encapsulation behavior.

Stack: Post-Quantum Cryptography, ML-KEM-768, X25519, Hybrid KEM, Zero-Trust, CPS Telemetry

[PROJECT / SEC-002] ZERO-TRUST PKI MESH

2025 / OPEN-SOURCE SECURITY REPOSITORY

Domain: Zero Trust / PKI / Infrastructure Security | Claim: Establishes a hardened internal multi-tier X.509 Public Key Infrastructure provisioning cryptography Internal X.509 Certificate Authority provisioning root, intermediate, and leaf certificates for edge Mutual TLS (mTLS) reverse proxy enforcement. Demonstrates multi-tier CA hierarchy management, certificate lifecycle tracking, and hardened Nginx reverse proxy configuration for mutual authentication.

Stack: Zero-Trust, PKI, mTLS, X.509, Nginx, Infrastructure Security

[PROJECT / INF-001] GITOPS MESH ORCHESTRATOR

2025 / OPEN-SOURCE INFRASTRUCTURE REPOSITORY

Domain: Distributed Systems / Infrastructure / Network Reliability | Claim: Orchestrates distributed service discovery and traffic shaping with dynamic TTL Service mesh orchestrator featuring dynamic TTL heartbeat leases, round-robin load distribution, token-bucket burst rate-limiting, and CIDR ingress inspection. Built for resilient service discovery and load balancing across multi-node topologies without centralized single-points-of-failure.

Stack: Service Mesh, GitOps, Load Balancing, Rate Limiting, Distributed Systems, Infrastructure

[PROJECT / SEC-004] ATTACK SURFACE INTELLIGENCE PIPELINE

2025 / OPEN-SOURCE SECURITY REPOSITORY

Domain: Cybersecurity / External Attack Surface Management / Security Automation | Claim: Automates external attack surface reconnaissance, normalization External Attack Surface Management (EASM) engine aggregating multi-source reconnaissance data, scoring exposure vectors, and outputting SIEM-ready JSON telemetry. Automates discovery of exposed infrastructure assets, domain names, certificate anomalies, and open network services.

Stack: EASM, Attack Surface, Threat Intelligence, Reconnaissance, SIEM, DevSecOps

[PROJECT / TEL-002] Automated SIEM & Telemetry Pipeline

2025 / ENTERPRISE DEPLOYMENT

Domain: Enterprise Infrastructure / Security Engineering | Claim: Establishes continuous host-level telemetry ingestion and automated security log auditing Engineered an automated telemetry and security monitoring pipeline using Wazuh HIDS/XDR to ingest host telemetry and audit logs. Configured real-time agent forwarding, kernel auditd integration, log normalization, and alert dispatching across multi-tier enterprise server environments.

Stack: Wazuh, HIDS / XDR, Linux Auditd, SIEM Telemetry, Log Auditing, Grafana

[PROJECT / SEC-003] Reverse-Proxy & Perimeter Hardening

2025 / PERIMETER SPECIFICATION

Domain: Enterprise Infrastructure / Security Engineering | Claim: Hardens web and network perimeters against unauthorized traffic through reverse-proxy Implemented robust reverse-proxy architectures using Nginx and Apache to isolate backend application nodes. Configured granular rate limiting, Web Application Firewall (WAF) rule sets, Certbot automated SSL/TLS provisioning, and TLS configuration tuning to enforce modern cryptographic standards and mitigate threat vectors.

Stack: Nginx, Apache, WAF, Rate Limiting, Certbot, SSL/TLS, Perimeter Hardening

[PROJECT / SYS-004] Infrastructure Automation & CI/CD Pipelines

2025 / AUTOMATION WORKFLOWS

Domain: Enterprise Infrastructure / Distributed Systems | Claim: Eliminates manual deployment drift and automates multi-environment synchronization using Architected automated deployment workflows and infrastructure synchronization pipelines using Bash, Python, and GitHub Actions. Automated secure SSH/FTP delivery, environment health validation, configuration templating, and operational maintenance routines to guarantee reliable application delivery.

Stack: CI/CD, GitHub Actions, Bash Scripting, Python, SSH Automation, Infrastructure

6. TECHNICAL NOTES & ENGINEERING MONOGRAPHS

- [NOTE / TN-001] Resonant Inductive Coupling: Efficiency, Frequency and Air-Gap Analysis"**

(Inductive Coupling & Resonance, TECHNICAL MONOGRAPH)
- [NOTE / TN-002] Distributed Network Telemetry and Intrusion Detection"**

(Telemetry & Distributed Systems, TECHNICAL MONOGRAPH)
- [NOTE / TN-003] SCADA Network Security Considerations for Power Infrastructure"**

(SCADA & CPS Security, TECHNICAL MONOGRAPH)

7. TECHNICAL COMPETENCIES

Power Systems & Hardware:

High-Q Resonant Inductive Coupling, Series-Series Tank Compensation, ZVS Inverters, Transmission Fault Studies, Digital Storage Oscilloscopes, Precision LCR Meters, Spectrum Analyzers.

Infrastructure, Security & Networking:

Linux Kernel Sockets (SO_RCVBUFFORCE), AF_PACKET MMAP Ring Buffers, SCADA Modbus-TCP / DNP3 Deep Packet Inspection, IEC 62443 Zone Segmentation, Ansible Automation, WireGuard Mesh.

8. VERIFIED CURRICULA & CREDENTIALS

Packt / Coursera: AWS Certified DevOps Engineer – Professional Specialization	[Verified Curriculum]
IBM: Introduction to Hardware and Operating Systems	[Verified Curriculum]
IBM: Introduction to Networking and Storage	[Verified Curriculum]
IBM: Introduction to Cybersecurity Essentials	[Verified Curriculum]
IBM: Cybersecurity Tools & Cyberattacks	[Verified Curriculum]
Northeastern University: Software Quality Control and Management Parts 1 & 2	[Verified Curriculum]
University of Leeds: Foundations of Software Testing and Validation	[Verified Curriculum]
Meta: Advanced Programming in Kotlin	[Verified Curriculum]
IBM: Introduction to Software, Programming, and Databases	[Verified Curriculum]
Skillshare / Coursera: Modern Frontend Web Development Specialization	[Verified Curriculum]

REFERENCES & INSTITUTIONAL VALIDATION:

Academic supervisor recommendations, authenticated degree transcripts, and institutional validation records available immediately upon request during formal evaluation.